

ECON104 Final Review

March 14, 2025

Logistics

- Exam is Tuesday 3-6PM. Exam is in Ackerman Union Grand Ballroom, so get there early so you can pick out a seat.
- Make sure you bring your student ID and a calculator that can do some scientific functions (logs, exponential function, etc.)
- "Feel" of the exam will be similar to the midterm:
 - 10 short answer questions, 10 multiple choice questions.
 - Recognize code and what it does.
- Some material from the midterm will be included (various tests, intuition, recognizing models), so refresh on the midterm content! Weight will be put mostly on the post-midterm material.

Panel Data

So far, we have dealt with two types of datasets:

- 1 Cross-Section: Many individuals at a single point in time
- 2 Time Series: One individual across many periods

The natural extension to these is the panel data setting, where we observe many individuals across many time periods.

Suppose we observe N individuals across T time periods were interested in estimating the following model:

$$y_{it} = \beta_1 + \beta_2 x_{it} + \beta_3 w_i + v_{it} \quad (1)$$

where observations are indexed by individual (i) and time period (t):

Pooled Regression

- The most basic panel regression is the "Pooled" regression.
- You simply estimate an OLS regression model treating each unit-period pair as its own observation
- For OLS to be BLUE:

$$\mathbb{E}(e_{it}) = 0 \quad (\text{Mean-Zero Error})$$

$$\text{Var}(e_{it}) = \sigma^2 \quad (\text{Homoskedasticity})$$

$$\text{Cov}(e_{is}, e_{jt}) = 0 \quad (\text{Independence})$$

$$\text{Cov}(e_{it}, x_{it}) = \text{Cov}(e_{it}, w_i) = 0 \quad (\text{Exogeneity})$$

Note that the independence is across individuals and time!

Cluster Robust Standard Errors

- Oftentimes, we think that the homoskedasticity is failed, but only across groups (i.e. clusters) of the sample.
 - Within a cluster, the homoskedasticity assumption is satisfied.
 - In many panel datasets, we assume individuals are clusters by themselves.
- In addition, we may think that independence is failed within clusters as well (i.e. my random error today is related to my random error yesterday).
- In settings where homoskedasticity and independence fail in this manner, OLS standard errors are no longer valid
 - Cluster robust standard errors as the solution

Fixed Effects

The original pooled regression assumes that everyone has the same intercept (β_1) and their errors are all pulled from a mean-zero distribution.

But this may not be true! In some situations, individuals may have persistent differences in outcome.

We can represent differences in these averages by giving everyone a different intercept (i.e. fixed effect):

$$y_{it} = \beta_{1i} + \beta_2 x_{it} + \beta_3 w_i + e_{it} \quad (2)$$

This can help solve issues of *bias* in your estimates of β_2 .

Dummy Variable OLS (DVLS)

One way to estimate this model is to plug in a full set of dummy variables D_i^j :

$$y_{it} = \sum_{j=1}^N \beta_{1j} D_i^j + \beta_2 x_{it} + \beta_3 w_i + e_{it} \quad (3)$$

where $D_i^j = 1$ when $j = i$ and zero otherwise.

- This guarantees errors are mean-zero across all individuals.
- Can help solve some (but not all) issues of endogeneity.
- You are estimating N new coefficients, so you need a very *tall* panel to get small standard errors.
- An F test with DVLS as the unrestricted regression and Pooled as the restricted regression is your test for fixed effects.

Fixed Effects

When N is very large and you can't directly estimate all the dummies, you can still estimate β_2 through either of the following methods:

① Difference Estimator

$$\begin{aligned}y_{it} &= \beta_1 i + \beta_2 x_{it} + \beta_3 w_i + e_{it} \\ -y_{it-1} &= \beta_1 i + \beta_2 x_{i,t-1} + \beta_3 w_i + e_{i,t-1} \\ \Delta y_{it} &= \beta_2 \Delta x_{it} + \Delta e_{i,t}\end{aligned}$$

② Within Estimator

$$\begin{aligned}y_{it} &= \beta_1 i + \beta_2 x_{it} + \beta_3 w_i + e_{it} \\ -\frac{1}{T} \sum_{t=1}^T y_{it} &= -\frac{1}{T} \sum_{t=1}^T (\beta_1 i + \beta_2 x_{i,t-1} + \beta_3 w_i + e_{i,t-1}) \\ \tilde{y}_{it} &= \beta_2 \tilde{x}_{it} + \tilde{e}_{it}\end{aligned}$$

Random Effects

If we believe the different intercepts are pulled independently of the covariates, then we instead would estimate a *random effects* model:

$$y_{it} = \bar{\beta}_1 + \beta_2 x_{it} + \beta_3 w_i + \underbrace{u_i + e_{it}}_{v_{it}} \quad (4)$$

Because the total error in a period v_{it} is autocorrelated through the persistence of u_i , we need to reweight the equation to get efficient estimators:

$$\underbrace{y_{it} - \theta \bar{y}_i}_{y_{it}^*} = \beta_1 \underbrace{(1 - \theta)}_{x_1^*} + \beta_2 \underbrace{(x_{it} - \theta \bar{x}_i)}_{x_{it}^*} + \beta_3 \underbrace{(1 - \theta) w_i}_{w_i^*} + v_{it}^* \quad (5)$$

Where

$$\theta = 1 - \frac{\sigma_e}{\sqrt{T\sigma_u^2 + \sigma_e^2}}$$

Some RE Comments

Consider our weight formula

$$\theta = 1 - \frac{\sigma_e}{\sqrt{T\sigma_u^2 + \sigma_e^2}}$$

1. When $\theta \rightarrow 0$, then RE $\rightarrow$ Pooled
 - Could happen in short panels (small T)
 - Generally happens when the random effect (σ_u^2) is small
2. When $\theta \rightarrow 1$, then RE $\rightarrow$ Within FE
 - Normally happens with very long panels (very large T)
 - You have enough data to see the persistent differences

Pooled vs Random Effects

- In the Random Effects model, we assume that autocorrelation is due to persistent heterogeneity.
- So if there is no autocorrelation, then there is no heterogeneity.
- Do a LM type test of autocorrelation:
 - Test statistic is distributed $N(0, 1)$
 - If autocorrelation, then $LM > z_{(1-\alpha)}$ and use RE
 - If no autocorrelation, then $LM < z_{(1-\alpha)}$ and use Pooled Regression

Fixed Effects vs Random Effects

How do we select between the two?

1. Random effects should only be used if the persistent error is unrelated to the set of RHS variables in the sample.
 - If there is endogeneity, then RE is biased but FE is unbiased
 - Do a different type of Hausman test:
 - Estimate the Fixed Effects model; pull b_{FE} and $\hat{\text{Var}}(b_{FE})$
 - Estimate the Random Effects model; pull b_{RE} and $\hat{\text{Var}}(b_{RE})$
 - Compute the test statistic:

$$t = \frac{b_{FE} - b_{RE}}{\sqrt{\hat{\text{Var}}(b_{FE}) - \hat{\text{Var}}(b_{RE})}}$$

- Rejecting the test tells you to use the Fixed Effects model, while failing to reject tells you to use the Random Effects model

Fixed Effects vs Random Effects

2. In the RE model, We assume that the selection of units is random (and large)
 - This interpretation drives the difference between random effect (u_i) and fixed effects (β_{1i})
 - In some panels, it makes sense to always use fixed effects (countries, states, large companies, etc.). The choice of units is inherently not random!
3. Fixed effect models cannot immediately estimate the coefficients of time-invariant variables:
 - Hausman-Taylor model is a type of IV approach applied to the RE model
 - Gets rid of endogeneity between the individual effect and whatever endogenous x or w you have.

Panel Test Examples: Pooled vs FE (Lab 6)

```
anova(FE.Full, FE.Pool)

## Analysis of Variance Table
##
## Model 1: log(cost) ~ log(output) + I(log(output)^2) + log(price) + load +
##      year + firm
## Model 2: log(cost) ~ log(output) + I(log(output)^2) + log(price) + load
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      66 0.17257
## 2      85 1.27492 -19   -1.1023 22.189 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 1: Using the 'lm' regression command

```
pFtest(fm_full, fm_no)

##
## F test for twoways effects
##
## data:  log(cost) ~ log(output) + I(log(output)^2) + log(price) + load
## F = 22.189, df1 = 19, df2 = 66, p-value < 2.2e-16
## alternative hypothesis: significant effects
```

Figure 2: Using the 'plm' regression command

Panel Test Examples: FE vs. RE(Lab 6)

```
phptest(fm_firm, fm_rfirm)
```

```
##
```

```
## Hausman Test
```

```
##
```

```
## data: log(cost) ~ log(output) + logoutput.sq + log(price) + load
```

```
## chisq = 5.6153, df = 4, p-value = 0.2298
```

```
## alternative hypothesis: one model is inconsistent
```

Dependent Variable Models

Our dependent variable models are used to determine the results of discrete (instead of continuous) outcomes.

- 1 Linear Probability Model (LPM):

$$Pr(Y_i = 1|X_i) = \beta_1 + \beta_2 X_i$$

- 2 Probit Model:

$$Pr(Y_i = 1|X_i) = \Phi(\beta_1 + \beta_2 X_i) \quad (\text{Standard Normal})$$

- 3 Logit Model:

$$Pr(Y_i = 1|X_i) = \Lambda(\beta_1 + \beta_2 X_i) = \frac{\exp(\beta_1 + \beta_2 X_i)}{1 + \exp(\beta_1 + \beta_2 X_i)}$$

Linear Probability Model

$$Pr(Y_i = 1|X_i) = \beta_1 + \beta_2 X_i$$

- Marginal Effects are always the same
- Probabilities can fall outside of 0 and 1
- Can be calculated with OLS

Probit Model

$$Pr(Y_i = 1|X_i; \beta) = \Phi(\beta_1 + \beta_2 X_i)$$

- Marginal Effects change with X :

$$\frac{dPr(Y_i = 1)}{dx_i} = \beta_2 \phi(\beta_1 + \beta_2 x_i)$$

- Probabilities are between 0 and 1
- Calculated with Maximum Likelihood Estimation:

$$LL(b) = \sum_{i=1}^N Y_i \log Pr(Y_i = 1|X_i; b) + (1 - Y_i) \log(Pr(Y_i = 0|X_i; b))$$

Probit Model

$$Pr(Y_i = 1|X_i) = \Phi(\beta_1 + \beta_2 X_i)$$

We can also report marginal effects differently:

- Average Marginal Effect (AME):

$$AME = \frac{1}{N} \sum_{i=1}^N \beta_2 \phi(\beta_1 + \beta_2 X_i)$$

- Marginal Effect at the Mean (MEM):

$$AME = \beta_2 \phi(\beta_1 + \beta_2 \bar{X})$$

Logit Model

$$Pr(Y_i = 1|X_i) = \Lambda(\beta_1 + \beta_2 X_i)$$

Notes:

- In binary case, logit is very similar to probit.
- Straightforward to calculate (very likely to show up on exam)
- Marginal Effects can be written in the following way:

$$\beta_2 \left(\frac{\exp[-(\beta_1 + \beta_2 X_i)]}{(1 + \exp[-(\beta_1 + \beta_2 X_i)])^2} \right)$$

$$\implies \beta_2 \Lambda(\beta_1 + \beta_2 X_i) [1 - \Lambda(\beta_1 + \beta_2 X_i)]$$

$$\implies \beta_2 Pr(Y = 1|X) [1 - Pr(Y = 1|X)]$$

Evaluating Binary Models

Oftentimes Binary models can be evaluated using a *Confusion Matrix*, which breaks down a into 4 categories:

Prediction	Reference	
	0	1
0	TN	FN
1	FP	TP

There are three main statistics used to evaluate models:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Wald Tests

When we want to test linear hypotheses with any maximum likelihood estimator, we use the Wald Test. However, the easiest way to think about a Wald Test is to just consider an F Test!

When estimating an OLS regression, the objective was to minimize the SSE in our sample. By applying some linear hypothesis (e.g. $\beta_2 = \beta_3$), we then create a *restricted model*. The F test statistic is then written as

$$F = \left(\frac{SSE_R}{SSE_U} - 1 \right) \frac{N - K}{J}$$

Focusing on the part in the parentheses, this is simply the percentage increase in *SSE* that your model is hit with in order to satisfy the restriction of the null hypothesis. In other words, it is a measure of how your model has been hurt by the restriction.

Wald Tests

Turning back to our MLE models, we are attempting to find parameter values that maximize the following likelihood:

$$L(\beta) = \prod_{i=1}^N \Pr(Y_i = y_i | x_i; \beta)$$

When we place some restriction on the parameters of the model, we have a similar result where the restricted model generates a smaller likelihood:

$$L_U \geq L_R$$

Wald Tests

The Wald test then takes the log-ratio of these to construct a test statistic based on the percentage difference between the unrestricted and restricted regressions:

$$2(\log L_U - \log L_R) \sim \chi^2_{(J)}$$

where the term inside the parentheses is the log-percentage increase in likelihood you gain from removing the restrictions of the null hypothesis.

Multinomial Models

Sometimes people have more than two options. In this case we used one of the following models:

- multinomial logit
- conditional logit
- multinomial logit

Multinomial Logit

Probability that i chooses j is given by

$$p_{ij} = \frac{\exp(\beta_{1j} + \beta_{2j}X_i)}{\sum_{k=1}^J \exp(\beta_{1k} + \beta_{2k}X_i)}$$

- Only includes person-specific attributes (X_i)
- We have to set some option to have $\beta_1 = 0$ and $\beta_2 = 0$
- Independence of Irrelevant Alternatives (IIA) holds:

$$\frac{p_{ij}}{p_{ik}} = \frac{\exp(\beta_{1j} + \beta_{2j}X_i)}{\exp(\beta_{1k} + \beta_{2k}X_i)}$$

This ratio does not depend on any other alternative choices.

Conditional Logit

Probability that i chooses j is given by

$$p_{ij} = \frac{\exp(\beta_1 j + \beta_2 X_{ij})}{\sum_{k=1}^J \exp(\beta_1 k + \beta_2 X_{ik})}$$

- Attributes are indexed by both individual and choice (X_{ij})
- We have to set some option to have $\beta_1 = 0$
- Essentially the same as MLM

Multinomial vs. Conditional Logit

These two models are essentially the same, so what is the difference?

→ How "preferences" to different options are considered.

Consider a situation where you are estimating a model of consumer choice using customer's disposable income (DI_i) and price of the goods (P_{ij}).

- Income is constant across choices, so we estimate different income effects for each good: β 's are different for each option
- Price is different across choices, but we have the same reaction to price regardless of good: β is fixed across objects.

Multinomial Probit

Probability is too complicated to write down (trust me)

- Normal distribution is used like the original probit model
- IIA does not hold
- Generally considered more flexible than MLM model

Other Models

- 1 **Ordered Models:** Either logit or probit can be adapted to cases where responses are discrete but ordered (sometimes called sentiment models).
- 2 **Count Data:** Count data use the poisson regression model:

$$E[Y] = \exp(\beta_1 + \beta_2 X)$$

- 3 **Censored Data:** If the observed variable has a lower (or upper) limit, then we need a **tobit** model to account for bias.

Ordered Models

Ordered models are used in situations where responses are discrete, but are ordered in some way (e.g. 1 to 5 star reviews, letter grades). Similar to the standard logit/probit models, we start with a linear model:

$$y^* = \underbrace{\beta_1 X_1 + \cdots + \beta_K X_K}_{\hat{y}} + \varepsilon$$

And outcomes are determined based on what interval y^* falls in:

$$Y = \begin{cases} M & \text{if } \mu_{M-1} \leq y^* \\ M-1 & \text{if } \mu_{M-2} \leq y^* < \mu_{M-1} \\ \vdots & \vdots \\ 2 & \text{if } \mu_1 \leq y^* < \mu_2 \\ 1 & \text{if } y^* < \mu_1 \end{cases}$$

Both the slope terms (β) and cutoffs (μ) are estimated

Ordered Models

Define $\hat{y} = \beta_1 X_1 + \dots + \beta_K X_K$

Probabilities of landing in any of the outcomes are determined by the CDF of the error terms:

$$Pr(Y = M) = 1 - Pr(\varepsilon \leq \mu_{M-1} - \hat{y})$$

$$Pr(Y = M - 1) = Pr(\varepsilon \leq \mu_{M-1} - \hat{y}) - Pr(\varepsilon \leq \mu_{M-2} - \hat{y})$$

$$\vdots \qquad \qquad \qquad \vdots$$

$$Pr(Y = 2) = Pr(\varepsilon \leq \mu_2 - \hat{y}) - Pr(\varepsilon \leq \mu_1 - \hat{y})$$

$$Pr(Y = 1) = Pr(\varepsilon \leq \mu_1 - \hat{y})$$

In the probit/logit models, these probabilities are replaced with $\Phi(\cdot)$ and $\Lambda(\cdot)$, respectively.

Example (Discussion Question 16.23b)

```
## Call:
## polr(formula = factor(hsat3) ~ age + I(age^2) + educ2 + working,
##       data = rwm88, method = "probit")
##
## Coefficients:
##               Value Std. Error t value
## age          -0.055709  0.0050842  -10.96
## I(age^2)      0.000408  0.0000725    5.62
## educ2         0.047848  0.0073468    6.51
## working       0.091071  0.0382128    2.38
##
## Intercepts:
##      Value      Std. Error t value
## 1|2   -1.253        0.001  -1124.772
## 2|3   -0.119        0.023    -5.175
```

Count Data

In situations where we face count data (e.g. number of cars in a family), we like to use the *Poisson Model*, which has the following results:

- Expectation

$$E[Y|X] = \exp(\beta_1 + \beta_2 X)$$

- Continuous Marginal Effect

$$\frac{\partial E[Y|X]}{\partial X} = \beta_2 E[Y|X]$$

- Dummy Variable Effect (with δ as the DV coefficient):

$$\text{Change} = \exp(\beta_1 + \beta_2 X + \delta) - \exp(\beta_1 + \beta_2 X)$$

$$\text{Ratio} = \frac{\exp(\beta_1 + \beta_2 X + \delta)}{\exp(\beta_1 + \beta_2 X)} = \exp(\delta)$$

Censored Data

Censored data refers to cases where the data has a limit (typically lower):

- Hours worked can't be negative
- Minimum wage gives the minimum earnings of workers

Like other models, we have an unobserved "true" outcome with linear model:

$$y^* = \beta_1 + \beta_2 X + \varepsilon$$

And the observed outcome is given by

$$Y = \begin{cases} 0 & \text{if } y^* \leq 0 \\ y^* & \text{if } y^* > 0 \end{cases}$$

Censored Data

Assuming errors are normally distributed (and exogeneity and homoskedasticity is satisfied), then we estimate by MLE:

$$L(\beta_1, \beta_2, \sigma) = \prod_{Y_i=0} \left(1 - \Phi \left(\frac{\beta_1 + \beta_2 X_i}{\sigma} \right) \right) \prod_{Y_i \geq 0} \phi \left[\frac{Y_i - \beta_1 - \beta_2 X_i}{\sigma} \right]$$

The marginal effect in this model is then given by:

$$\frac{\partial E[Y|X]}{\delta X} = \beta_2 \Phi \left(\frac{\beta_1 + \beta_2 X}{\sigma} \right)$$

This is essentially the "true" marginal effect, β_2 , weighted by the probability that you observe the "true" outcome y^* , $\Phi(\cdot)$.

Statistical Learning

In general, the error of a model can be broken down into three categories:

$$\text{Model Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

- **Bias:** Refers to whether your prediction is correct on average.
- **Variance:** Refers to how large your prediction errors are on average.
- **Irreducible:** Refers to the variance inherent in the model (think variance of population errors). We can't fix this part!

Note that all of these refer to the **outcome** and not the **parameters** (y vs. β).

Bias-Variance Tradeoff

Typically there is a tradeoff between bias and variance:

- Adding in new variables reduces bias, but you increase variance as you overfit the data
- Removing variables prevents you from overfitting the data, but you increase the bias of your results

The simplest method of fixing this is picking the most informative subset of variables

- Can be computed using AIC/BIC like we did in Project 1
- Possibly an extremely difficult process if there are too many variables (too many subsets to check)

Regularization

One other alternative is adding a penalty to a normal OLS calculation:

- 1 **Ridge:** Penalize the sum of square parameters:

$$\beta^{Ridge} = \arg \min \sum_{i=1}^N (Y_i - b_0 - b_1 x_{1i} - \dots - b_K x_{1K})^2 + \lambda \sum_{i=1}^K b_k^2$$

- 2 **Lasso:** Penalize the sum of absolute values:

$$\beta^{LASSO} = \arg \min \sum_{i=1}^N (Y_i - b_0 - b_1 x_{1i} - \dots - b_K x_{1K})^2 + \lambda \sum_{i=1}^K |b_k|$$

Ridge vs. Lasso

There are plusses and minuses to each approach:

- Ridge has an exact solution, so you can derive some properties related to bias and variance.
- Lasso has no "closed-form" solution, but computers can calculate it quicker in many cases (mostly when K and N are very large).
- Lasso can fix parameter values to zero, while ridge cannot.
- Ridge performs well when very large parameter estimates are causing issues.

Elastic Net

Ridge and Lasso can be combined into a single estimator:

$$\beta^{EN} = \arg \min SSE + \lambda_R \sum_{i=1}^K b_k^2 + \lambda_L \sum_{i=1}^K |b_k|$$

Oftentimes we write the penalties in the following way:

$$\lambda = \lambda_L + \lambda_R$$

$$\lambda_L = \alpha \lambda$$

$$\lambda_R = (1 - \alpha) \lambda$$